

Exam

Copyright reserved

Electricity and Electronics EBN 111

7 July 2021

Test Information

Maximum marks:	93	Full marks:	90
Duration of test:	200 min	Open/Closed book:	Open
Total number of pages (this page included)		33	

Lecturers: Dr D. le Roux, Dr S. Smith, Prof X. Ye**Assistant Lecturers:** L. Kokot, M. Maritz, J. Thiselton, M. Trivedi, D. Wepener**Important**

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers, including complex numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your semester test, you declare the following:**

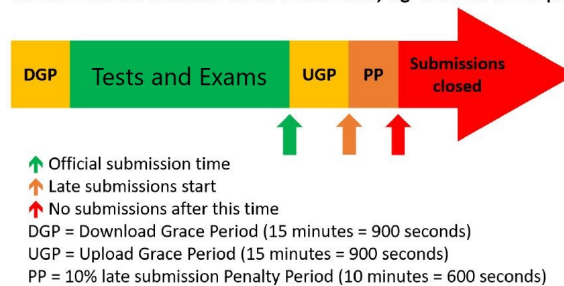
The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all tests for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the test for students to download all documents related to the test and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the test. Students must work through the test at an average rate of at least 2 minutes per mark.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the test.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.ams@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final test mark.
8. No late submissions are accepted after the PP.

Late submissions of session-based assessments, e.g. tests and online practicals



Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct:** $1-j2$ or $-3+j4$
 - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
 - Capital J is however incorrect, and $1-2j$ is also not acceptable.
 - If a complex number has no real part, for example $0+j2$, write $0+j2$ as the final answer.
- **Correct:**
 - With angle in degrees: $12\angle 34\deg$;
 - With angle in radians: $12\angle 0.5934$
 - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect:**
 - $1-i2$ (Use j instead of i)
 - $1 - j2$ (no spaces between characters)
 - $12 \angle 34 \deg$; (no spaces between characters)
 - $j10+1$ (Real part before the imaginary part for rectangular form.)
 - $-3+4j$ (should be $j4$ instead of $4j$)

Section 1

Questions 1 to 2 [4]

Given: I , x .

- | | |
|-----------|--|
| 01 | A cellphone battery provides 3.024 kJ for 24 hours to a cellphone at a constant current of I mA. Determine the voltage of the battery V_s . [2] |
| 02 | The current entering the positive terminal of an element is $i(t) = 10e^{-xt}$ mA, while the voltage across the element is given by $v(t) = 5e^{-xt}$ V. Calculate the energy associated with the element w_{element} over the interval 0 s to 4 s. [2] |

* $I = 4.5 \text{ mA}$.

01 $w = \int p dt$

$$3024 \text{ J} = V_s \times I \times 24 \text{ h} \times 3600 \text{ s/h}$$

$$V_s = \frac{3024}{24 \times 3600 \times 4.5 \times 10^{-3}}$$

$$= \underline{7.778 \text{ V}}$$

* $x = 0.25$

02 $w = \int_0^4 p dt = \int_0^4 v \times i dt$

$$= \int_0^4 5 \exp(-xt) \times 10 \exp(-xt) \times 10^{-3} dt$$

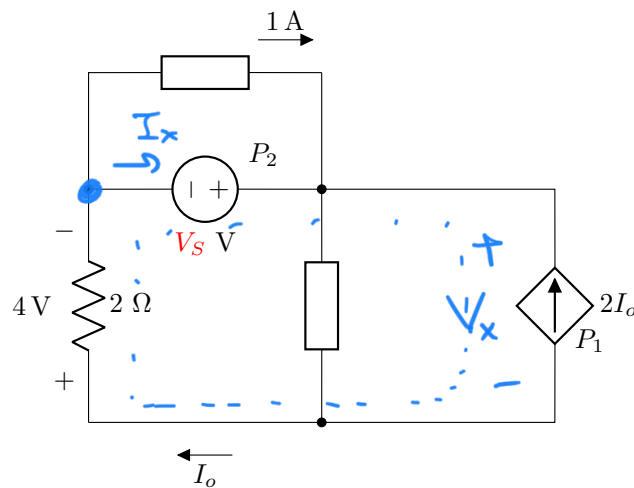
$$= 0.05 \int_0^4 \exp(-2xt) dt$$

$$= \frac{-0.05}{2x} e^{-2xt} \Big|_0^4$$

$$= \frac{-0.05}{2(0.25)} \left[e^{-2 \times 0.25 \times 4} - e^0 \right]$$

$$= \underline{86.467 \text{ mJ}}$$

Questions 3 to 4 [3]

Given: V_s .Find the power associated with the dependent current source P_1 and the power associated with the independent voltage source P_2 .03 P_1 [2]04 P_2 [1]

$$\ast V_s = 1V$$

$$\underline{03} \ast KVL: +4 - 1 + V_x = 0$$

$$\therefore V_x = -3V$$

$$\ast I_o = \frac{4}{2} = 2A.$$

$$\ast P_1 = -(V_x)(2I_o)$$

$$= -(-3)(2 \times 2)$$

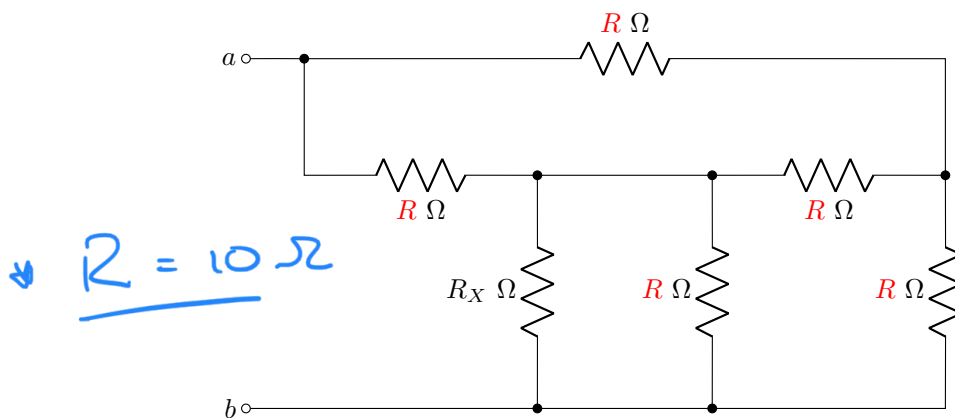
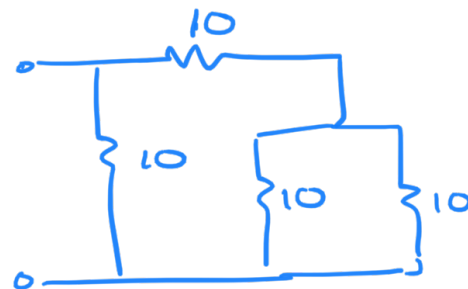
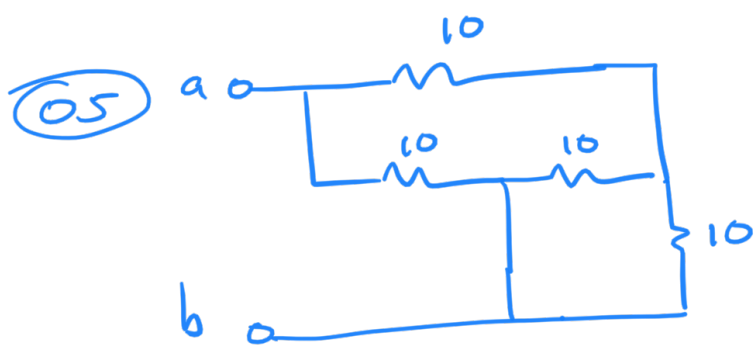
$$= \underline{12W}$$

$$\underline{04} \ast KCL: I_x = I_o - 1 = 1A.$$

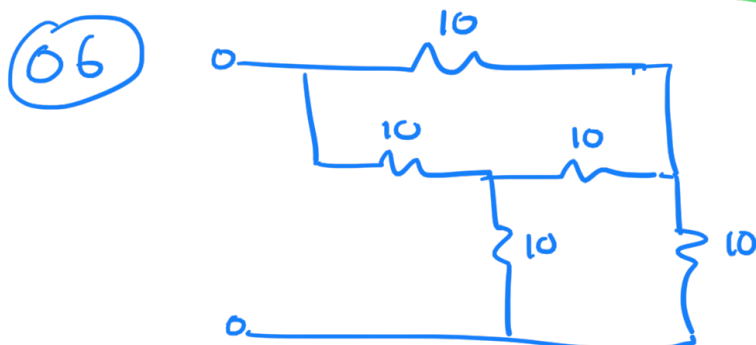
$$\ast P_2 = -(1V)(1A)$$

$$= \underline{-1W}$$

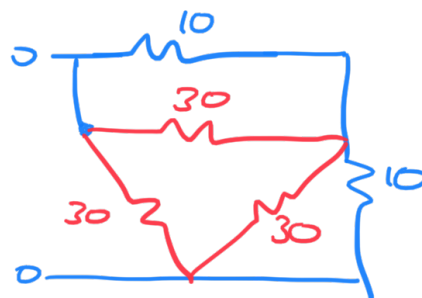
Questions 5 to 6 [4]

Given: R Determine the equivalent resistance R_{ab} for the case where $R_X = 0 \Omega$ and for the case where $R_X \rightarrow \infty \Omega$.05 R_{ab} if $R_X = 0 \Omega$ [2]06 R_{ab} if $R_X \rightarrow \infty \Omega$ [2]

$$\begin{aligned}
 R_{ab} &= 10 \parallel (10 + 10 \parallel 10) \\
 &= 10 \parallel (10 + 5) = \frac{10 \times 15}{10 + 15} \\
 &= 6 \Omega
 \end{aligned}$$

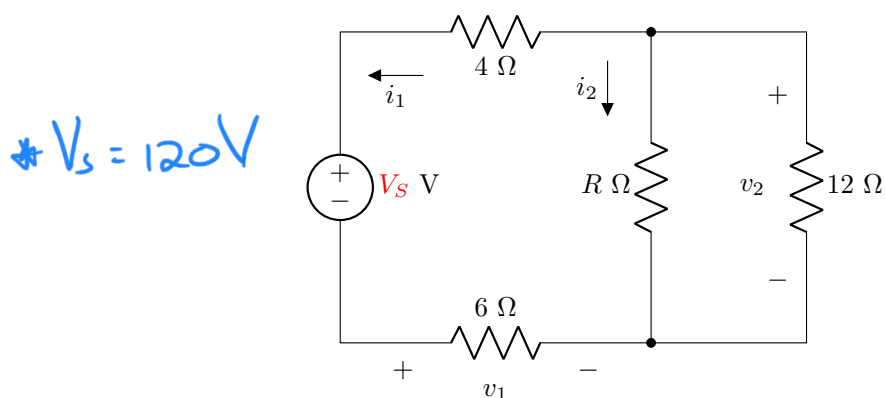


$$R_{\Delta} = 3R$$



$$\begin{aligned}
 R_{ab} &= 30 \parallel (10 \parallel 30 + 10 \parallel 30) \\
 &= 30 \parallel (7.5 + 7.5) \\
 &= 10 \Omega
 \end{aligned}$$

Questions 7 to 10 [4]

Given: V_S 

- | | |
|----|--|
| 07 | Determine i_1 if $R = 0 \Omega$ [1] |
| 08 | Determine v_1 if $R = 60 \Omega$ [1] |
| 09 | Determine i_2 if $R = 60 \Omega$ [1] |
| 10 | Determine v_2 if $R \rightarrow \infty \Omega$ [1] |

07

$$i_1 = \frac{-120}{4+6} = -12A \rightarrow$$

08

$$12 \parallel 60 = 10 \Omega$$

$$v_1 = \frac{-(120) \times 6}{10+4+6} = -36V \rightarrow$$

09

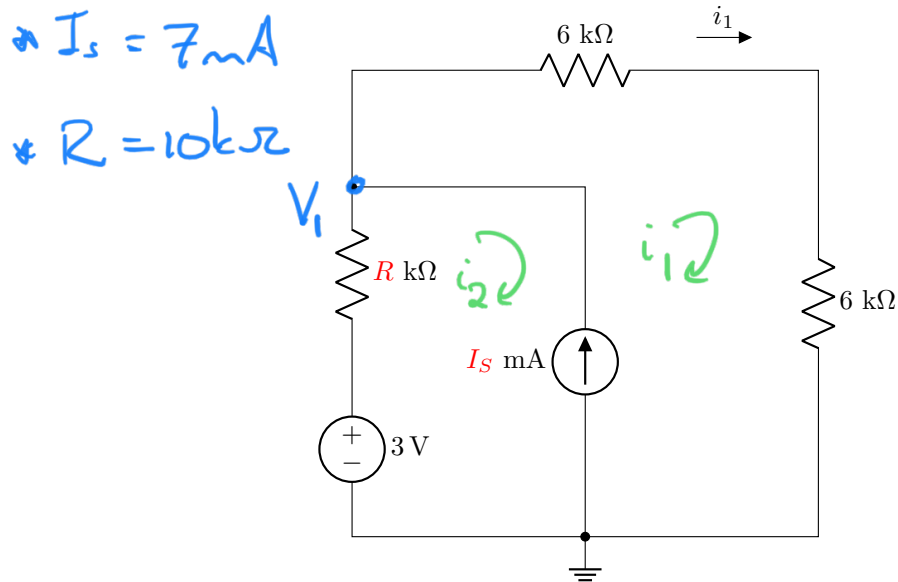
$$V_{11} = \frac{120 \times (60 \parallel 12)}{4+60 \parallel 12+6} = 60V \rightarrow$$

$$\therefore i_2 = \frac{V_{11}}{60} = 1A \rightarrow$$

10

$$v_2 = \frac{120 \times 12}{4+12+6} = 65.454V \rightarrow$$

Question 11 [2]

Given: I_S , R .11 Use either nodal or mesh analysis to determine i_1 . [2]

Nodal

$$7 \text{ mA} = \frac{V_1 - 3}{10 \text{ k}} + \frac{V_1}{12 \text{ k}}$$

$$(\times 120 \text{ k}) \quad 840 = 12V_1 - 36 + 10V_1$$

$$V_1 = \frac{840 + 36}{22} = 39.818 \text{ V}$$

$$i_1 = \frac{V_1}{12 \text{ k}} = \underline{3.318 \text{ mA}}$$

Mesh

$$i_1 - i_2 = 7 \text{ mA} \Rightarrow i_2 = i_1 - 7 \text{ mA}$$

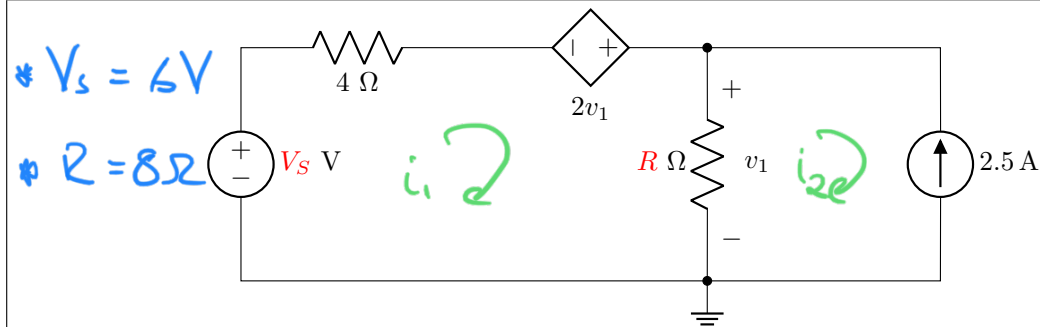
$$-3 + 10 \text{ k} i_2 + 12 \text{ k} i_1 = 0$$

$$-3 + 10 \text{ k} (i_1 - 7 \text{ mA}) + 12 \text{ k} i_1 = 0$$

$$22 \text{ k} i_1 = 3 + 70$$

$$i_1 = \underline{3.318 \text{ mA}}$$

Question 12 [2]

Given: V_S, R .12 Use either nodal or mesh analysis to determine v_1 . [2]

Nodal

$$\frac{v_1}{8} + \frac{v_1 - 2v_1 - 6}{4} = 2.5$$

($\times 8$) $v_1 + 2(v_1 - 2v_1) - 12 = 20$

$$-v_1 = 32$$

$$v_1 = \underline{\underline{-32V}}$$

Mesh

$$i_2 = -2.5A$$

$$v_1 = 8(i_1 - i_2)$$

$$v_1 = 8i_1 - 8(-2.5) = 8i_1 + 20$$

$$-6 + (4+8)i_1 - 8(-2.5) - 2v_1 = 0$$

$$-6 + 12i_1 + 20 - 2v_1 = 0$$

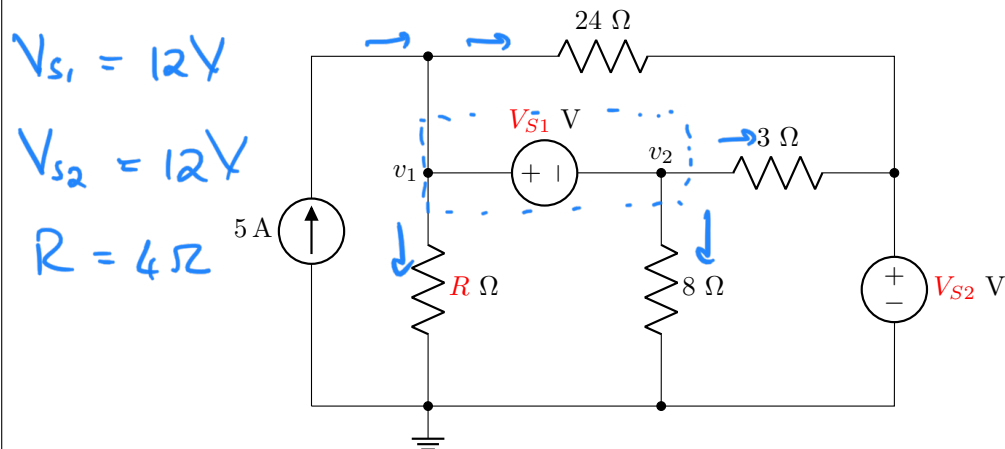
$$14 + 12\left(\frac{v_1 - 20}{8}\right) - 2v_1 = 0$$

$$14 + \frac{12v_1}{8} - \frac{20 \times 12}{8} - 2v_1 = 0$$

$$(1.5 - 2)v_1 = 30 - 14$$

$$v_1 = \frac{16}{-0.5} = \underline{\underline{-32V}}$$

Questions 13 to 16 [4]

Given: V_{S1} , V_{S2} , R .

Use nodal analysis to set up the following two equations of the form:

$$av_1 + 11v_2 = b$$

$$1v_1 + cv_2 = d$$

Calculate the unknown coefficients a , b , c and d .

13	a [1]	14	b [1]	15	c [1]	16	d [1]
----	---------	----	---------	----	---------	----	---------

* Supernode:

$$5 = \frac{v_1}{4} + \frac{v_1 - 12}{24} + \frac{v_2 - 12}{3} + \frac{v_2}{8}$$

$$(\times 24) \quad 120 = 6v_1 + v_1 - 12 + 8v_2 - 96 + 3v_2$$

$$\underline{7v_1 + 11v_2 = 228} \rightarrow$$

* KVL at V_{S1} :

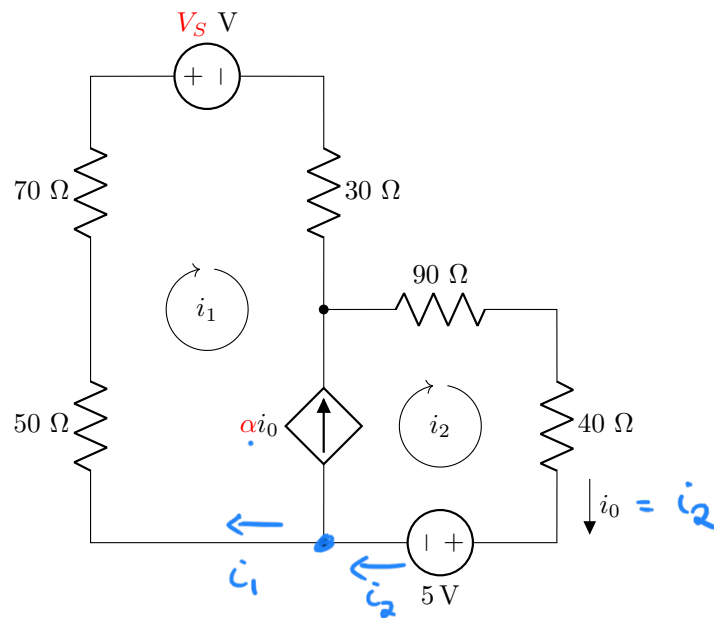
$$\underline{v_1 - v_2 = 12} \rightarrow$$

Questions 17 to 20 [4]

Given: α , V_S .

$$\alpha = 2$$

$$V_S = 11 \text{ V}$$



Use mesh analysis to set up the following two equations of the form:

$$150i_1 + ei_2 = f$$

$$-1i_1 + gi_2 = h$$

Calculate the unknown coefficients e , f , g and h .

17	e [1]	18	f [1]	19	g [1]	20	h [1]
----	---------	----	---------	----	---------	----	---------

Supermesh:
 ① + ②

$$(50 + 70 + 30)i_1 + (90 + 40)i_2 = -11 - 5$$

$$150i_1 + 130i_2 = -16$$

KCL:

$$i_2 = i_1 + 2i_0 \quad \text{but } i_0 = i_2$$

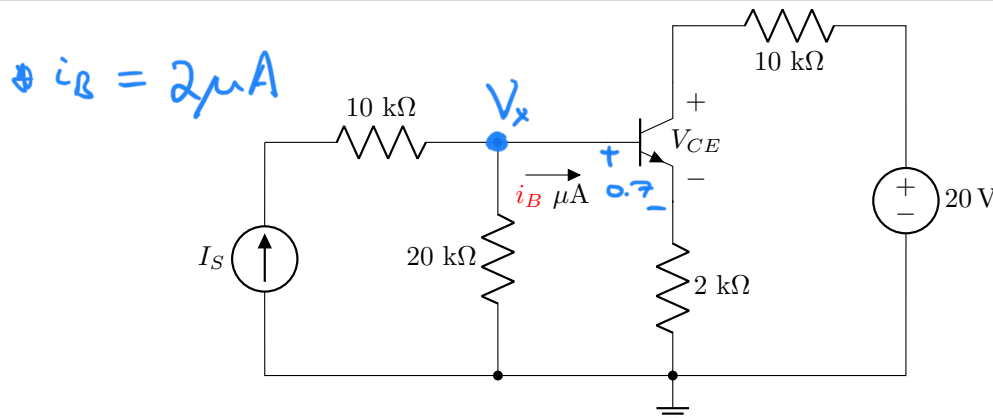
$$-i_1 + i_2 - 2i_2 = 0$$

$$\therefore -i_1 - i_2 = 0$$

Questions 21 to 22 [4]

Given: i_B

It is known that $\beta = 99$ and $V_{BE} = 0.7$ V. Given the value of the base current i_B , determine the source current I_S and the collector-emitter voltage V_{CE} .

21 I_S [2]22 V_{CE} [2]

21 KCL: $I_S = \frac{V_x}{20k} + 2 \mu A$

but $V_x = 0.7 + 2k(i_E)$
 $= 0.7 + 2k(99+1)i_B$
 $= 0.7 + 2k \times 100 \times 2 \mu$
 $= 1.1 V$

$\therefore I_S = \frac{1.1}{20k} + 2 \mu = \underline{57 \mu A} \rightarrow$

22 $-20 + 10k i_C + V_{CE} + 2k i_E = 0$

$V_{CE} = 20 - 10k(99)i_B - 2k(99+1)i_B$
 $= \underline{17.62 V} \rightarrow$

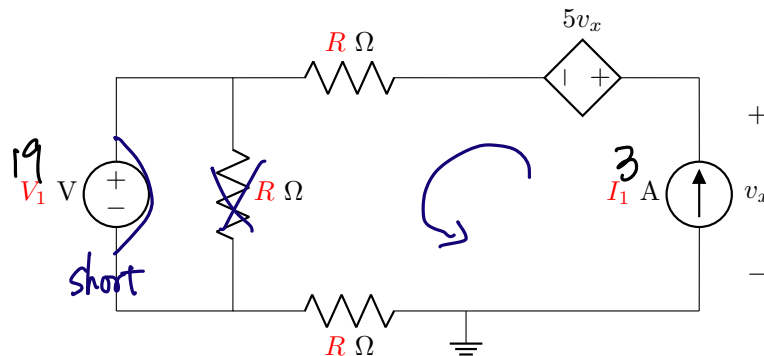
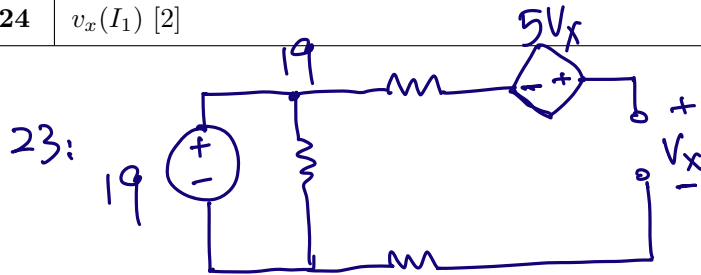
Total Section 1: [31]

Section 2

Questions 23 to 24 [4]

Given: I_1 , V_1 , R . **70**

Determine $v_x(V_1)$, the independent contribution of V_1 to v_x , and $v_x(I_1)$, the independent contribution of I_1 to v_x .

23 $v_x(V_1)$ [2]24 $v_x(I_1)$ [2]

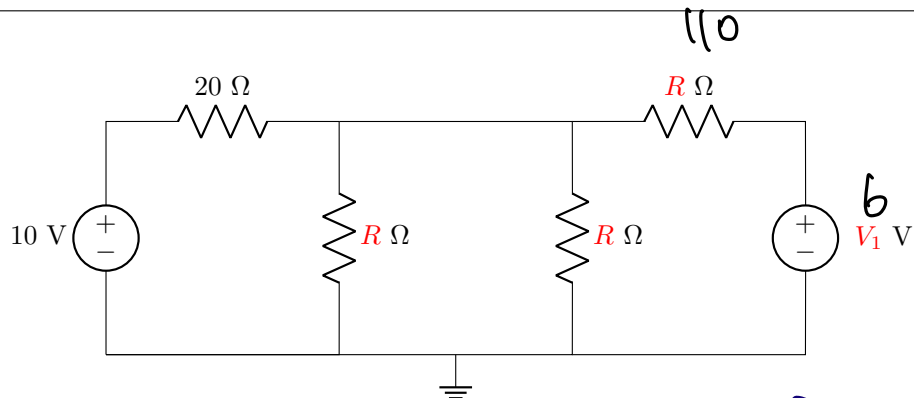
$$\begin{aligned}
 -19 - 5V_x + V_x &= 0 \\
 \Rightarrow 4V_x &= -19 \\
 \Rightarrow V_x &= 4.75V.
 \end{aligned}$$

$$\begin{aligned}
 24: \quad 5V_x + 3R + 3R - V_x &= 0 \quad (\text{KVL on original circuit}) \\
 \Rightarrow 4V_x &= -6R \Rightarrow V_x = -1.5R = -105V.
 \end{aligned}$$

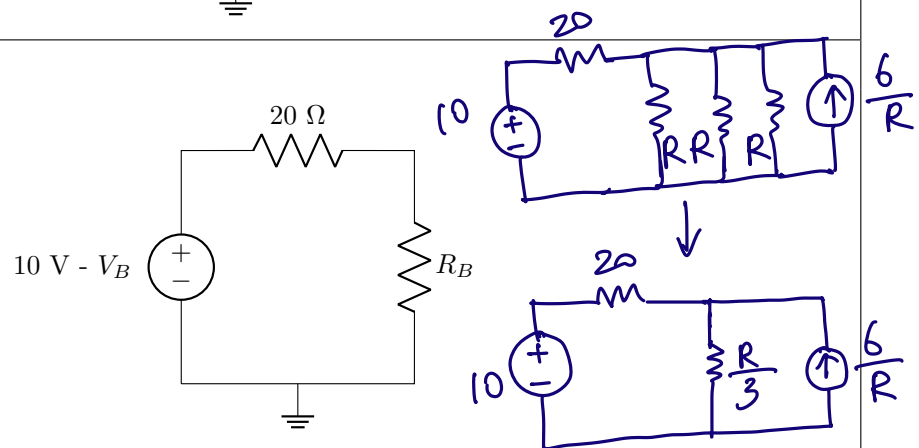
Questions 25 to 26 [3]

Given: V_1 , R .Use source transformation to obtain Circuit B so that it is equivalent to Circuit A, and determine R_B and V_B in Circuit B.

Circuit A:

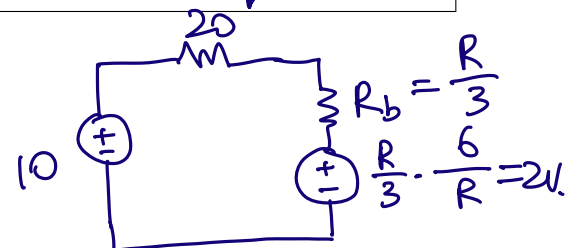


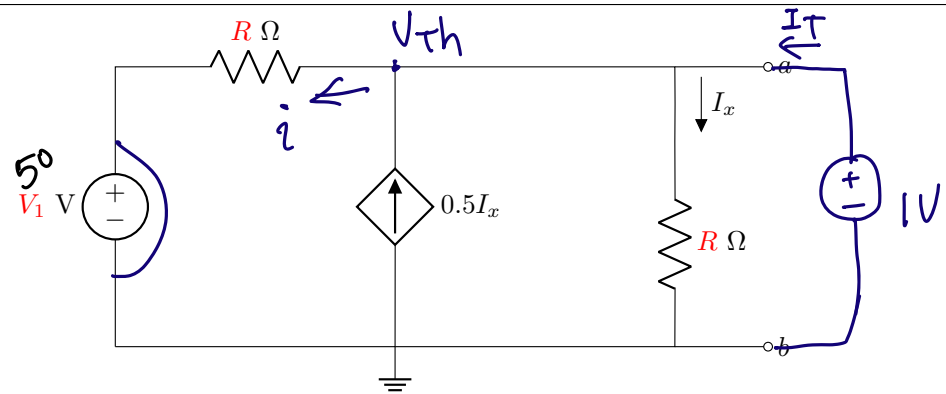
Circuit B:

25 R_B [1]26 V_B [2]

$$V_B = 2\text{ V}$$

$$R_B = \frac{R}{3} = 36.667\ \Omega.$$



Questions 27 to 28 [4]	
Given: $V_1, R, 350$	
Determine the Thevenin equivalent voltage V_{Th} and resistance R_{Th} between terminals $a-b$.	
	
27	V_{Th} [2]
28	R_{Th} [2]

$$27: \frac{V_{Th} - 50}{R} - 0.5I_x + \frac{V_{Th}}{R} = 0$$

$$\frac{V_{Th} - 50}{R} + 0.5 \frac{V_{Th}}{R} = 0$$

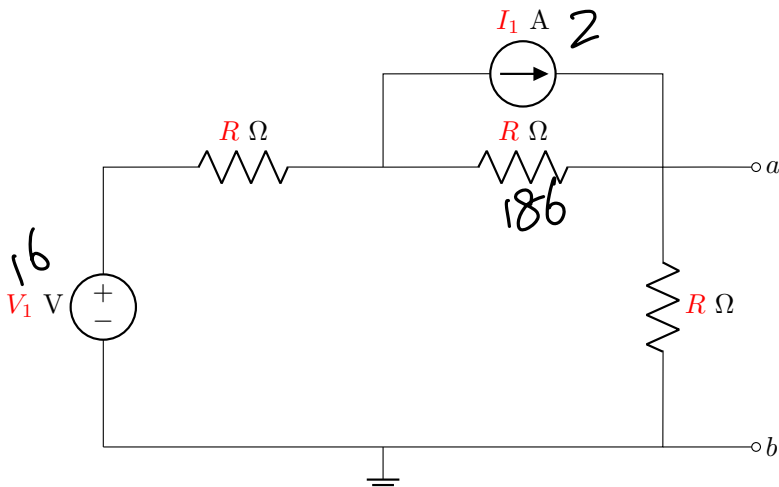
$$\frac{1.5 V_{Th}}{R} = \frac{50}{R} \Rightarrow 1.5 V_{Th} = 50 \Rightarrow V_{Th} = 33.333 \text{ V.}$$

$$28: I_x = \frac{1}{R}, \quad i = \frac{1}{R}$$

$$\Rightarrow i - 0.5I_x + I_x - I_T = 0$$

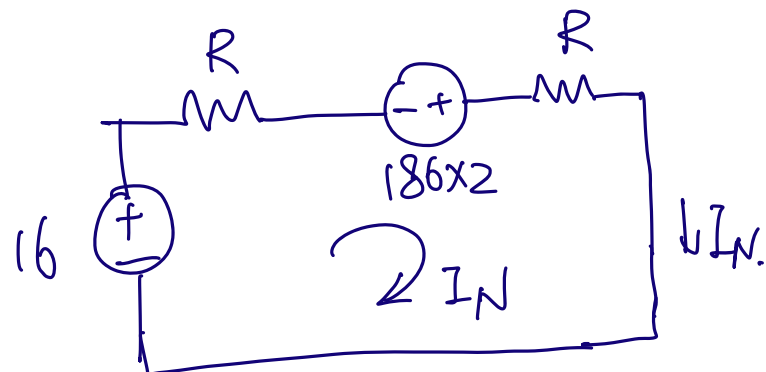
$$I_T = 0.5I_x + i = \frac{1.5}{R}$$

$$\Rightarrow R_{Th} = \frac{1}{I_T} = \frac{R}{1.5} = 233.33 \Omega.$$

Questions 29 to 30 [4]	
Given: V_1 , I_1 , R .	
Determine the Norton equivalent current I_N and resistance R_N between terminals a - b .	
	
29	I_N [2]
30	R_N [2]

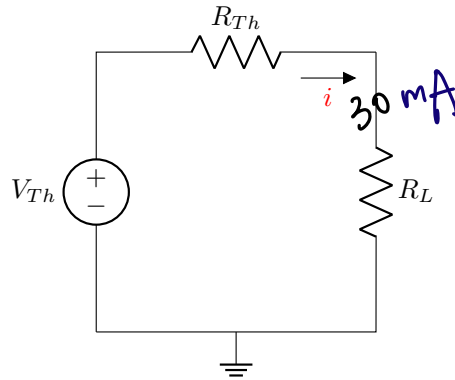
$$R_N = R \parallel 2R = \frac{R \times 2R}{R + 2R} = \frac{2}{3}R = 124 \Omega.$$

$$I_N = \frac{16 + 186 \times 2}{2R} = 1.0430 \text{ A.}$$



Question 31 [2]Given: i , P_{max} . $\leftarrow 35 \text{ mW}$.

The Thevenin equivalent circuit below is connected to a load resistor R_L . The maximum power P_{max} delivered to R_L is given. Determine the Thevenin equivalent voltage V_{Th} .

31 | V_{Th} [2]

$$i = \frac{V_{Th}}{2R_{Th}} = 30 \text{ mA}, \quad V_{Th} = 60 \times 10^{-3} R_{Th}$$

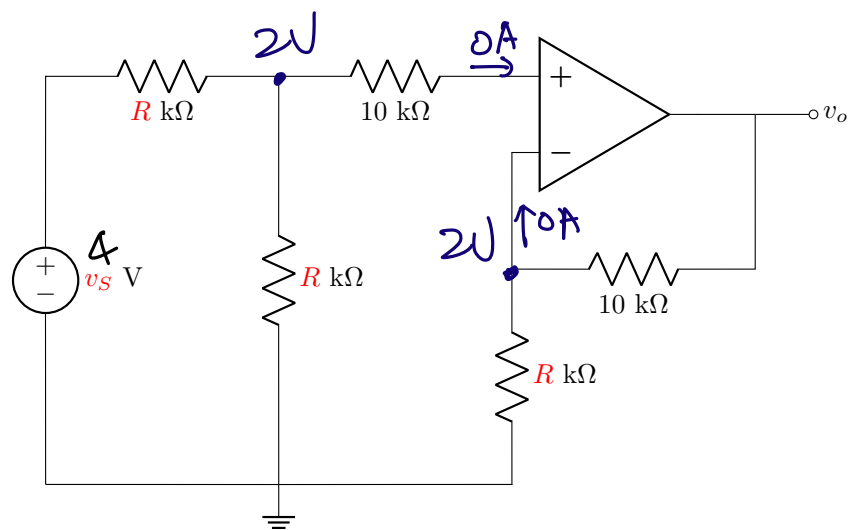
$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = 35 \text{ mW}$$

$$\frac{(60 \times 10^{-3})^2 R_{Th}}{4 R_{Th}} = 35 \times 10^{-3}$$

$$(60 \times 10^{-3})^2 R_{Th} = 4 \times 35 \times 10^{-3}$$

$$\Rightarrow R_{Th} = \frac{4 \times 35}{60^2 \times 10^{-3}} = \frac{350}{9} \Omega$$

$$\Rightarrow V_{Th} = 60 \times 10^{-3} \times \frac{350}{9} = 2.3333 \text{ V.}$$

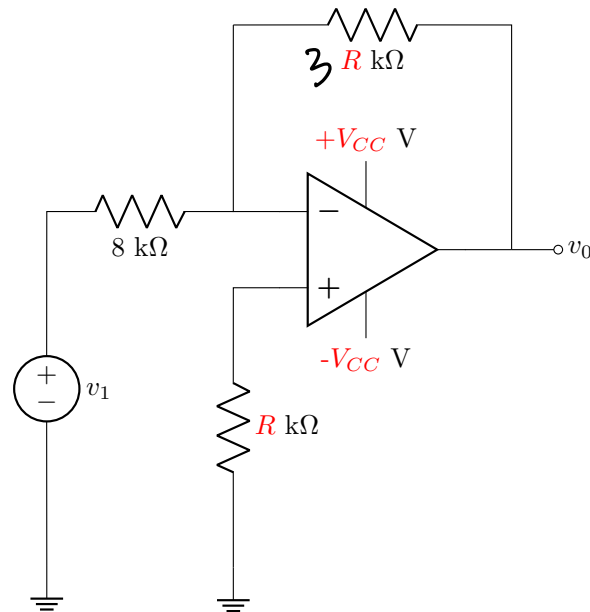
Question 32 [2]Given: v_S , R , q Assume that the op-amp is ideal. Determine the voltage v_o .**32** v_o [2]

$$2V = V_o \cdot \frac{R}{10+R}$$

$$\Rightarrow V_o = \frac{10+q}{q} \times 2 = 4.2222 \text{ V.}$$

Question 33 [2]Given: R, V_{CC} .

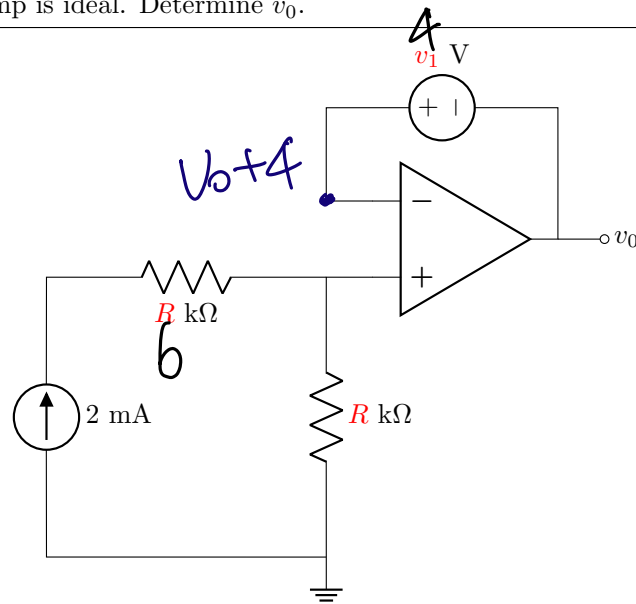
Assume that the op-amp is ideal. Determine the minimum value of v_1 before the op-amp goes into positive saturation.

33 v_1 [2]

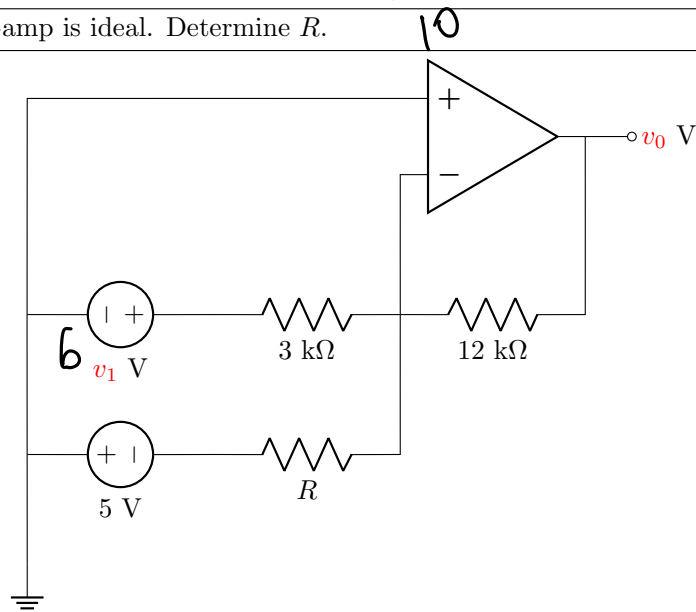
$$v_o = -\frac{3}{8} v_1 \leq 10$$

$$\Rightarrow v_1 \geq -\frac{80}{3}$$

$$\min(v_1) = \frac{80}{3} = -26.667 \text{ V.}$$

Question 34 [2]Given: v_1 , R .Assume that the op-amp is ideal. Determine v_0 .**34** | v_0 [2]

$$v_0 + 4 = 2 \times 6 = 12$$
$$\Rightarrow v_0 = 8 \text{ V.}$$

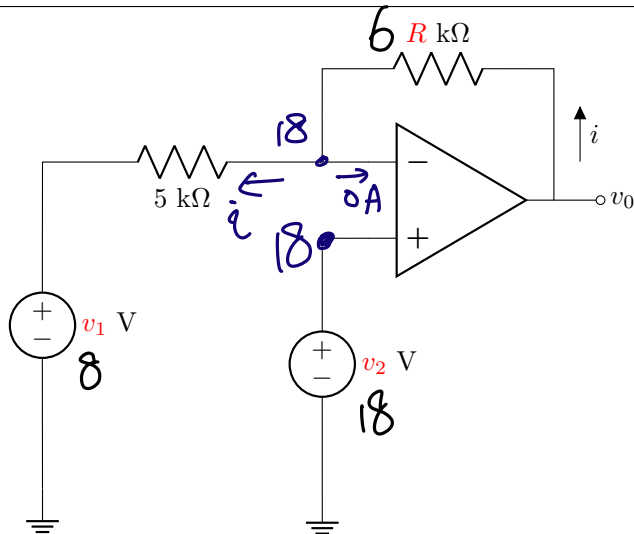
Question 35 [2]Given: v_1 , v_0 .Assume that the op-amp is ideal. Determine R .35 | R [2]

$$v_0 = -v_1 \cdot \frac{12}{3} + [-(-5)] \cdot \frac{12}{R} = 10$$

$$= -6 \cdot \frac{12}{3} + 5 \cdot \frac{12}{R} = 10$$

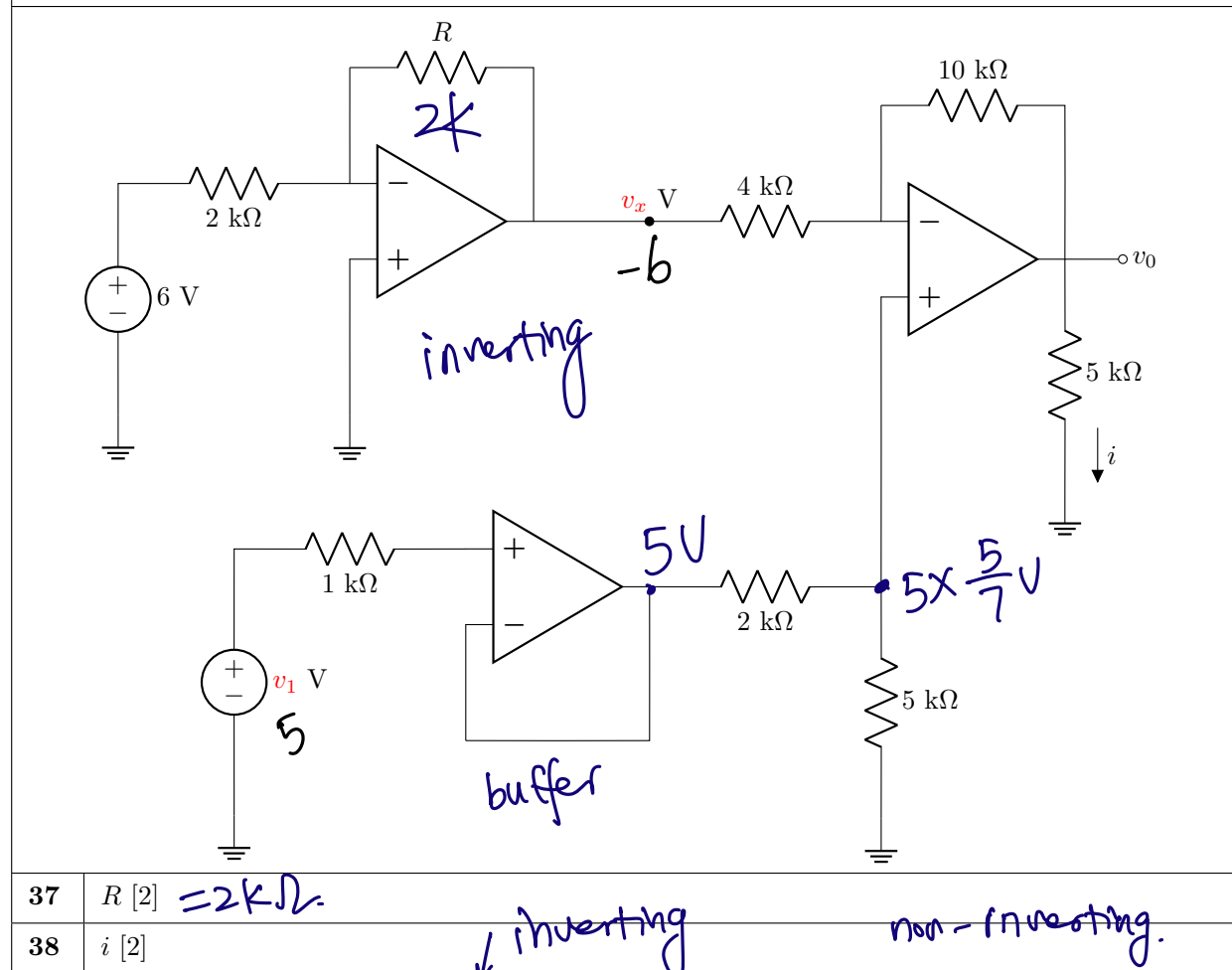
$$\Rightarrow \frac{60}{R} = 34$$

$$\Rightarrow R = 1.7647 \text{ k}\Omega.$$

Question 36 [2]Given: v_1 , v_2 , R .Assume that the op-amp is ideal. Determine i .36 i [2]

$$i = \frac{v_o - 18}{6k} = \frac{18 - 8}{5k} = 2 \text{ mA}$$

Questions 37 to 38 [4]

Given: v_1 , v_x .Assume that the op-amps are ideal. Calculate the resistor R and the current i .

$$v_0 = -(-6) \cdot \frac{10}{4} + 5 \times \frac{5}{7} \left(1 + \frac{10}{4}\right)$$

$$= 15 + 12.5 = 27.5$$

$$i = \frac{v_0}{5} = 5.5 \text{ mA}$$

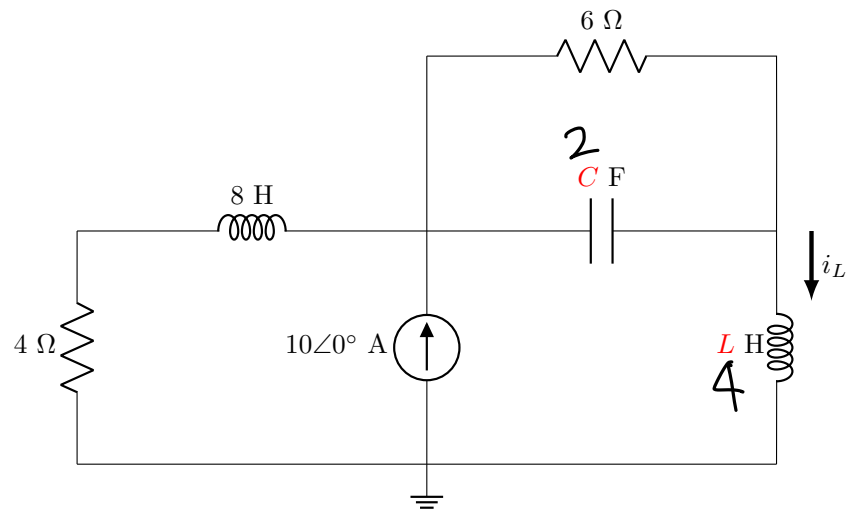
Total Section 2: [31]

Section 3

Questions 39 to 40 [3]

Given: C , L

The circuit below is under steady DC conditions.
--



39	Find the energy w_C stored in the capacitor. [2]
-----------	---

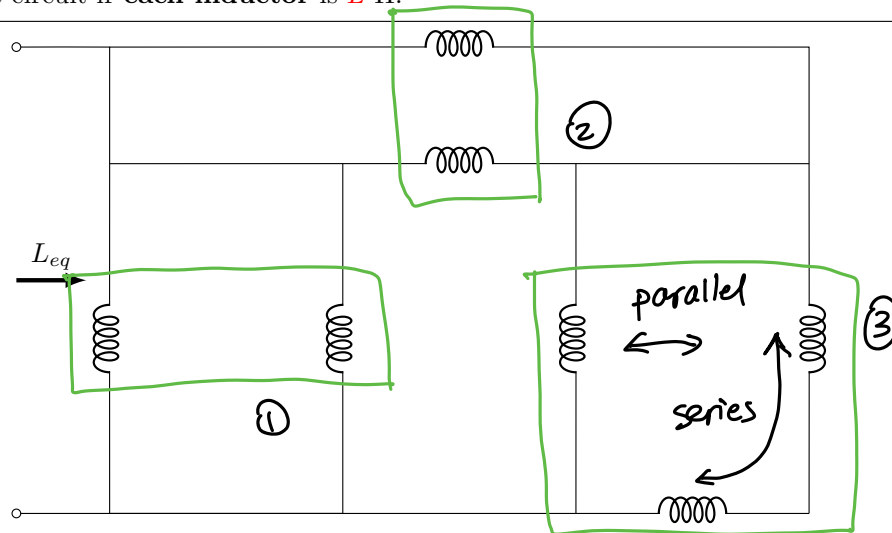
40	Find the current i_L through the inductor. [1]
-----------	---

$$40: i_L = 10 \cdot \frac{4}{4+6} = 4 \text{ A}$$

$$V_C = i_L \cdot 6 = 24 \text{ V}$$

$$39: w_C = \frac{1}{2} C V^2 = \frac{1}{2} \times 2 \times 24^2 = 576 \text{ J}$$

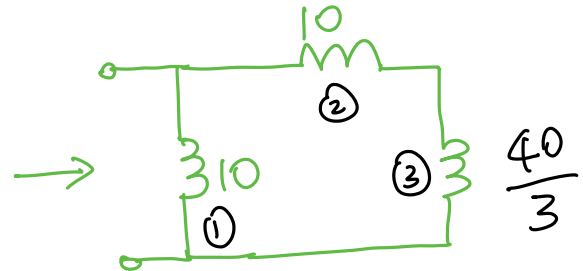
Question 41 [2]

Given: L . 20Find L_{eq} in the circuit if each inductor is L H.

$$\frac{20 \times 40}{60} = \frac{40}{3}$$

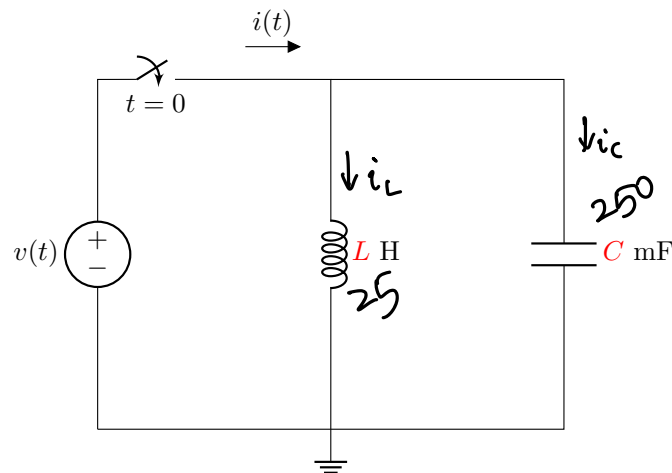
41 L_{eq} [2]

$$\begin{aligned}
 L_{eq} &= 10 \parallel \left(10 + \frac{40}{3}\right) \\
 &= \frac{10 \times \frac{70}{3}}{10 + \frac{70}{3}} \\
 &= \frac{700}{100} = 7 \text{ H.}
 \end{aligned}$$



Question 42 [2]Given: C, L .

In the circuit below, the inductor and capacitor are initially uncharged as the switch will only be closed at $t = 0$. Find $i(t)$ at $t=2$ s when $v(t) = 2t + 1$ V.

42 | $i(2 \text{ s})$ [2]

$$i(t) = i_L + i_C = \frac{1}{L} \int_0^t (2t+1) dt + C \frac{dv}{dt}$$

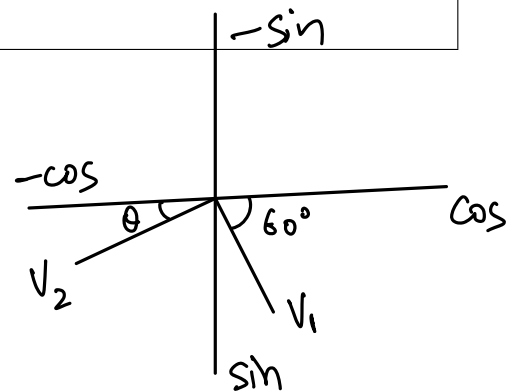
$$= \frac{1}{25} [t^2 + t]_0^t + 0.25 \cdot 2$$

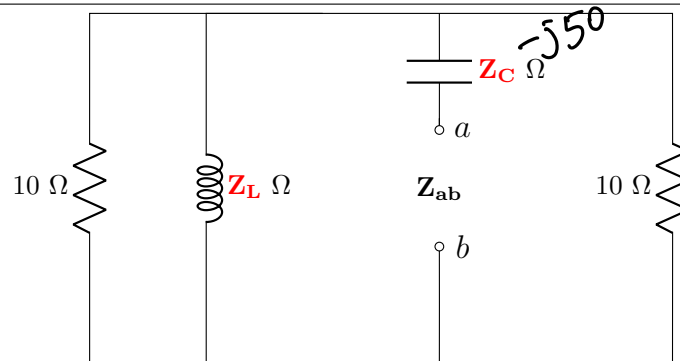
$$= \frac{1}{25} (t^2 + t) + 0.5$$

$$i(2) = \frac{1}{25} \times 6 + 0.5 = 0.74 \text{ A.}$$

Questions 43 to 44 [4]	
Given: A, θ	
Consider two sinusoids $v_1(t)$ and $v_2(t)$, where	
$v_1(t) = 5 \sin(\omega t + 30^\circ) \text{ V and}$ $v_2(t) = -2 \cos(\omega t + \theta^\circ) \text{ V, } \theta^\circ \in [0^\circ, 30^\circ].$	
The sinusoids $v_1(t)$ and $v_2(t)$ are represented by phasors $\mathbf{V}_1$ and $\mathbf{V}_2$, respectively.	
43	Choose one correct statement from the following list. ✓ 1. v_1 leads v_2 by $(120-\theta)^\circ$. 2. v_2 leads v_1 by $(120-\theta)^\circ$. 3. v_1 leads v_2 by $(90-\theta)^\circ$. 4. v_2 leads v_1 by $(90-\theta)^\circ$. Write your answer as i if statement i is correct, i=1, 2, 3, and 4. [2]
44	Find $\mathbf{V}_T = \mathbf{V}_1 + \mathbf{V}_2$ in polar form . [2]

$$\begin{aligned}
 V_T &= 5 \angle -60^\circ + 2 \angle -152^\circ \\
 &= 5.3200 \angle -82.068^\circ
 \end{aligned}$$



Question 45 [2]Given: Z_C , Z_L $j5$ Find the equivalent impedance Z_{ab} in **rectangular form** at terminals a - b .45 Z_{ab} (rectangular form) [2]

$$Z_1 = 10 || 10 = 5$$

$$Z_2 = Z_1 || Z_L = 5 || j5 = \frac{5 \times j5}{5 + j5} = 2.5 + j2.5$$

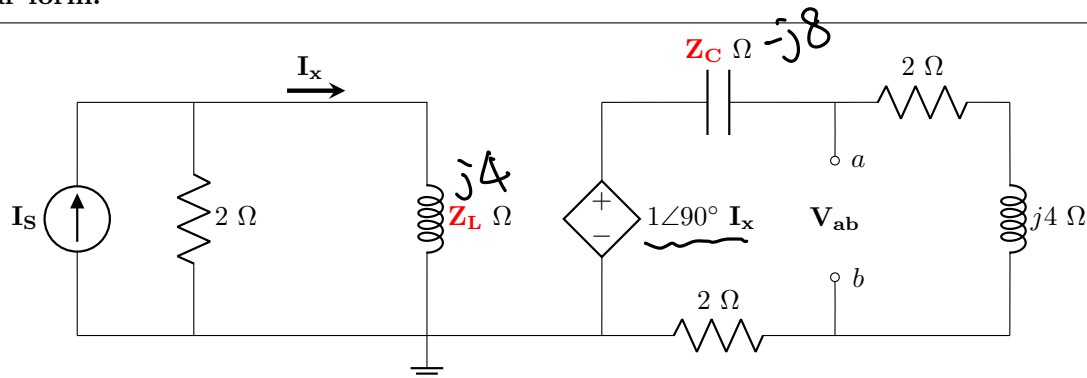
$$Z_{ab} = Z_C + Z_2 = -j50 + 2.5 + j2.5$$

$$= 2.5 - j47.5 \Omega.$$

Questions 46 to 47 [4]

Given: Z_C , Z_L

If $I_x = 2\angle 0^\circ$ A, find the phasor current I_S and phasor voltage V_{ab} . Write your answers in **rectangular form**.



46 I_S (rectangular form). [2]

47 V_{ab} (rectangular form). [2]

$$46: \quad I_x = 2 = I_S \cdot \frac{2}{2+j4}$$

$$\Rightarrow I_S = 2+j4 \text{ A}$$

$$47: \quad 1\angle 90^\circ I_x = j \cdot 2 = j2$$

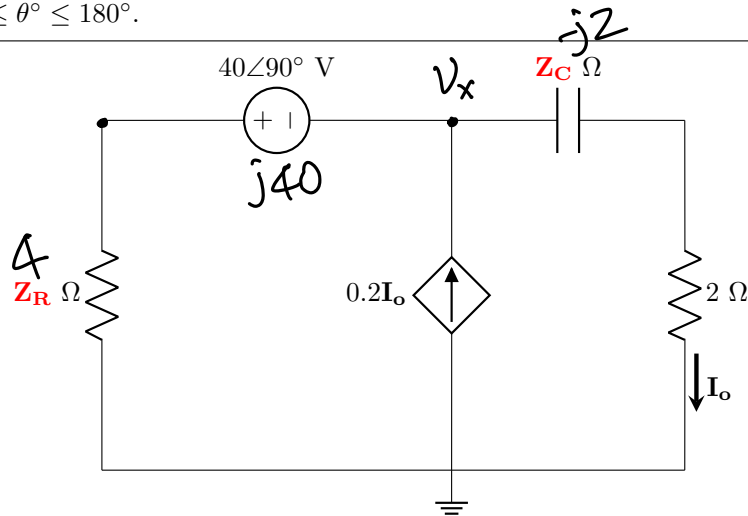
$$V_{ab} = j2 \cdot \frac{2+j4}{2+2+j4-j8} = j2 \cdot \frac{2+j4}{4-j4}$$

$$= -1.5 - j0.5 \text{ V}$$

Question 48 [2]

Given: Z_R , Z_C

Use nodal analysis to find the phasor current $I_o = r\angle\theta^\circ$ A. Write your answers in **polar form**, where $r \geq 0$ and $-180^\circ \leq \theta^\circ \leq 180^\circ$.

48 I_o (polar form) [2]

$$\frac{V_x + j40}{4} - 0.2I_o + I_o = 0, \quad I_o = \frac{V_x}{2 - j2}$$

$$\frac{V_x + j40}{4} = -0.8I_o = -0.8 \frac{V_x}{2 - j2}$$

$$\left(\frac{1}{4} + \frac{0.8}{2 - j2}\right)V_x = -j10$$

$$V_x = -\frac{j10}{0.45 + j0.2}, \quad I_o = \frac{V_x}{2 - j2}$$

$$I_o = -\frac{j10}{(0.45 + j0.2)(2 - j2)}$$

$$= 7.1796 \angle -68.962^\circ \text{ A}$$

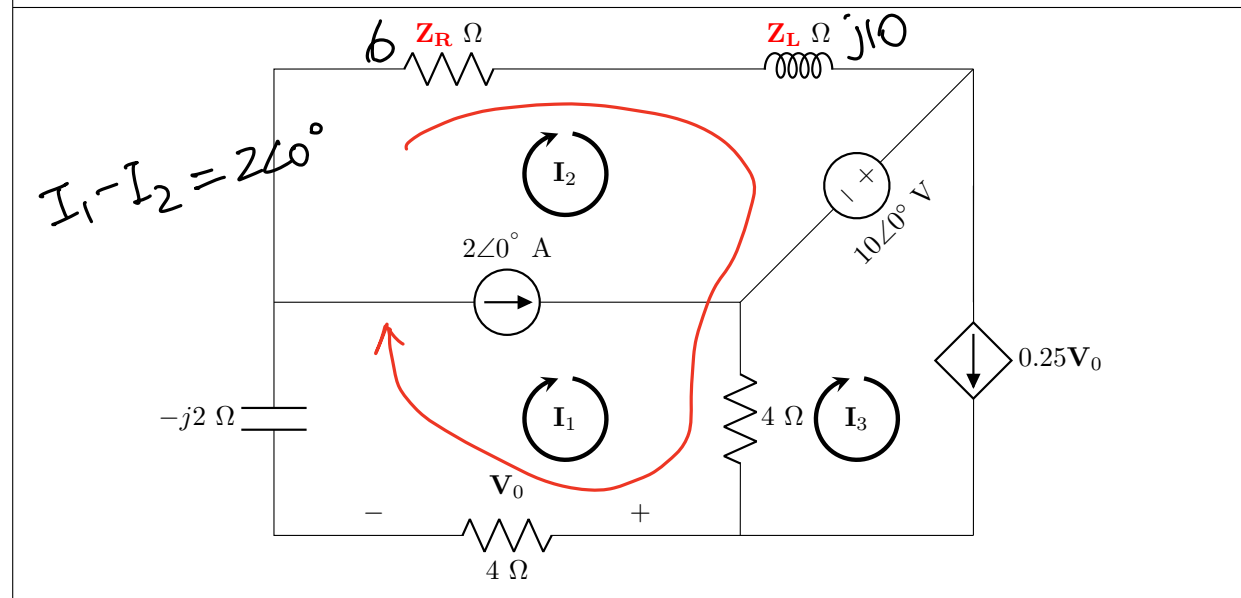
Questions 49 to 52 [4]

Given: $\mathbf{Z_R}$, $\mathbf{Z_L}$

Use mesh analysis to setup two equations of the following form for the circuit below:

$$\mathbf{aI_1} + \mathbf{bI_2} = 2\angle 0^\circ,$$

$$(4 - j2)\mathbf{I_1} + \mathbf{cI_2} = \mathbf{d}.$$

Find the unknown coefficients $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{d}$ in **rectangular form**.

49	$\mathbf{a}$ [1]	$1 + j0$	50	$\mathbf{b}$ [1]	$-1 + j0$	51	$\mathbf{c}$ [1]	$6 + j10$	52	$\mathbf{d}$ [1]	$-10 + j0$
----	------------------	----------	----	------------------	-----------	----	------------------	-----------	----	------------------	------------

$$I_3 = 0.25V_0 = 0.25 \times 4 I_1 = I_1$$

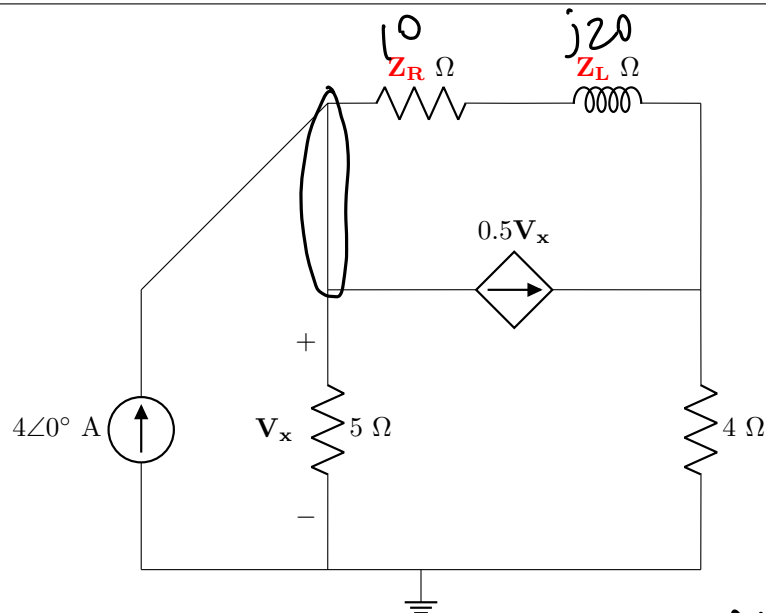
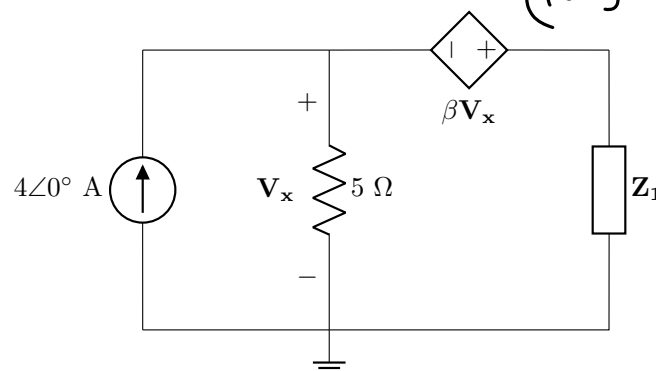
$$-j2 I_1 + (6 + j10) I_2 + 10 + 4(I_1 - I_3) + 4I_1 = 0$$

$$(4 - j2) I_1 + (6 + j10) I_2 = -10.$$

Questions 53 to 54 [2]

Given: $\mathbf{Z_R}$, $\mathbf{Z_L}$

Use source transformation to transform *Circuit A*, such that it is equivalent to *Circuit B*. In *Circuit B*, find the impedance $\mathbf{Z_1}$ and the unknown coefficient β .

Circuit A:*Circuit B:*

$$(10 + j20) \times 0.5 = 5 + j10$$

β

$$10 + j20 + 4 = 14 + j20$$

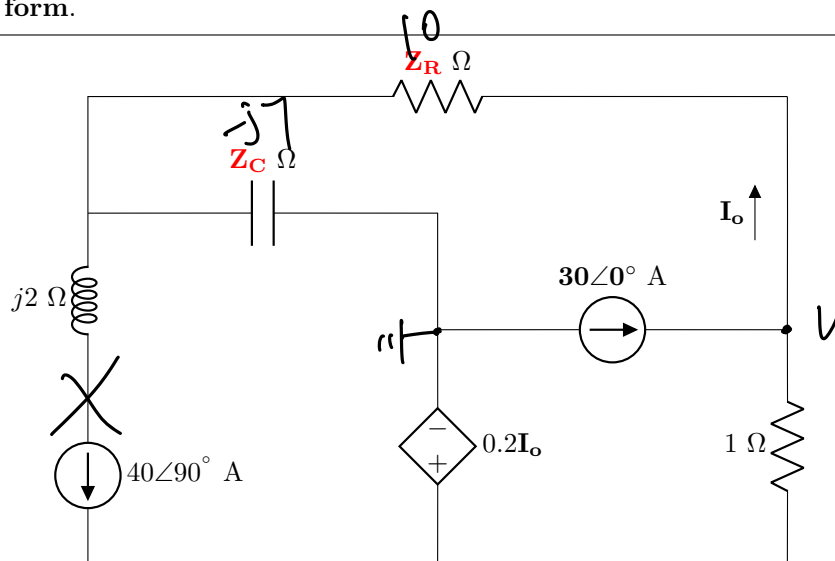
53 In *Circuit B*, find the impedance $\mathbf{Z_1}$ in rectangular form. [1]

54 In *Circuit B*, find the coefficient β in rectangular form. [1]

Question 55 [2]

Given: Z_R , Z_C

Determine I_o^A , the exclusive contribution of the $30\angle 0^\circ$ A current source to I_o . Write your answer in rectangular form.

55 I_o^A (rectangular form) [2]

$$-30 + \frac{V}{10-j7} + V - 0.2I_o = 0$$

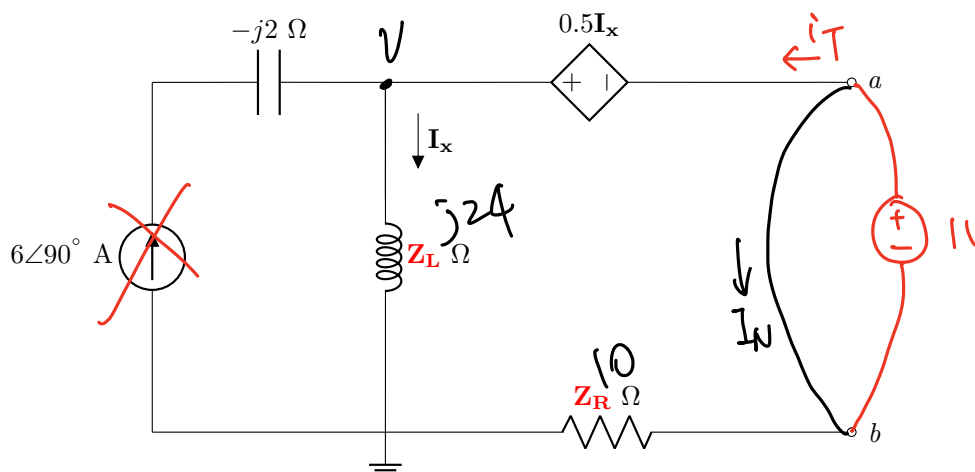
$$\frac{V}{10-j7} + V - 0.2 \frac{V}{10-j7} = 30$$

$$\left(\frac{0.8}{10-j7} + 1\right)V = 30$$

$$V = \frac{30(10-j7)}{10.8-j7}$$

$$I_o = \frac{V}{10-j7} = \frac{30}{10.8-j7} = 1.9560 + j1.2678 \text{ A.}$$

Questions 56 to 57 [4]

Given: Z_R , Z_L Find the Norton equivalent current I_N and impedance Z_N at terminals a - b . Write your answers in rectangular form.56 I_N (rectangular form) [2]57 Z_N (rectangular form) [2]

$$-j6 + \frac{V}{j24} + \frac{V - 0.5I_x}{10} = 0$$

$$\frac{V}{j24} + \frac{V - 0.5 \frac{V}{j24}}{10} = j6$$

$$\left(\frac{1}{j24} + \frac{1}{10} - \frac{1}{j480} \right) V = j6$$

$$V = \frac{j6}{\frac{1}{j24} + \frac{1}{10} - \frac{1}{j480}}$$

$$I_N = j6 - \frac{V}{j24} = j6 - \frac{j6}{1 + \frac{j24}{10} - \frac{1}{20}} = -2.1614 + j5.1445 \text{ A}$$

$$i_T = I_x$$

$$-0.5I_x + (10 + j24)I_x - 1 = 0$$

$$\Rightarrow (9.5 + j24)I_x = 1$$

$$I_x = \frac{1}{9.5 + j24} \text{ A}$$

$$Z_N = \frac{1}{I_x} = 9.5 + j24 \Omega$$

Total Section 3: [31]

Grand Total : [93]